MD. Rafi Imam

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GRE: 304 (Q: 164) | **IELTS Academic:** 7.0

RESEARCH INTERESTS

Parameter-Efficient Fine-Tuning (PEFT) • Trustworthy Robust Machine Learning • Representation Uncertainty Quantification • Multimodal Foundation Models • Efficient AI Systems Model Adaptation • Computer Vision Document AI

EDUCATION

Rajshahi University of Engineering & Technology (RUET) 2018 – 2023
B.Sc. in Electrical & Computer Engineering **CGPA: 3.30/4.00**
Last 60 credit hours GPA: 3.55/4.00
Relevant Coursework: Machine Learning, Neural Network, Natural Language Processing, Image Processing, Data Structures, Algorithms, Digital Signal Processing, Computer Networks
Standardized Tests: GRE 304 (Quantitative: **164**, Verbal: 140, AWA: 3.0) | IELTS Academic Overall **7.0** (Listening: 7.0, Reading: 7.0, Writing: 6.5, Speaking: 6.5)

PUBLICATIONS

1. **R. Imam**, et al., *Suicidal Thought Detection Using NLP on Reddit Data*, IEEE ICCIT 2023.
2. Md. M. Hossain, et al., **R. Imam**, *Privacy-Preserving Federated Learning for Lung Cancer Classification*, IEEE ICCIT 2023.
3. T. H. Mou, et al., **R. Imam**, *Detecting Polycystic Ovary Syndrome Using Machine Learning*, IEEE ICCIT 2023.
4. Z. Afrin, et al., **R. Imam**, *YOLOv8-based Object Detection for Self-driving Cars*, IEEE ICCIT 2023.

INDEPENDENT RESEARCH PROJECTS

GateConcat: Learned Token Gating for Robust Parameter-Efficient Fine-Tuning August 2026

- Developed GateConcat, a learned token-wise gate that estimates token quality from hidden-state deviations ($\Delta = h_{LoRA} - h_{base}$) and concatenates it with LoRA representations for robust classification under OCR corruption.
- Under 70% character substitution and OCR confusion, GateConcat improved noisy macro F1 over standard LoRA by $+0.0151$ ($p = 0.003$) and $+0.0230$ ($p = 0.002$), with high stability across seeds.
- Established that explicit corruption detection (AUC 0.91) yields zero downstream F1 benefit, while the unassisted gate (AUC ≈ 0.52) acts as an implicit label-salience feature—decoupling noise detection from task performance.

SALLoRA: Diagnostic Study on Robust PEFT July 2026

- Designed and evaluated SALLoRA, a robust PEFT framework combining token-level quality weighting, layer-wise scalars, and gradient sensitivity masking.
- Under strictly controlled conditions, demonstrated that static quality weighting does not reliably improve weighted F1 and degrades macro F1 by up to -0.0309, serving as a cautionary case study on the fragility of robustness claims in PEFT literature.

RESEARCH & PROFESSIONAL EXPERIENCE

Software Development Officer (AI & Automation) Dec 2023 – Present
Jamuna Bank PLC, AI & Automation Division

- **ML Infrastructure:** Designed and deployed a real-time Face Liveness & Presentation Attack Detection System, decoupling heavy PyTorch CUDA inferences (FastAPI) from stateless asynchronous gateway orchestrators (`asyncio.gather`), reducing evaluation latency from >5.0s to ˜800ms.
- **SALLoRA Online Updates:** Integrated client-side SALLoRA background workers to execute 2-epoch online micro-batch updates via a security API, enabling hot-reloading of calibrated model weights with zero system downtime.
- **Distributed Pipelines:** Architected event-driven, high-concurrency data streaming pipelines using Apache Kafka and Docker to ingest and process transaction-triggered risk flags.
- Applied **Swin-B forward hooks** for intermediate feature extraction within the AI-integrated financial systems, supporting OCR pipelines, face matching, and structured data extraction for real-world decision workflows.
- Developed **data processing and validation pipelines** involving AML checks, deduplication, and transaction profiling for risk-aware systems.
- Applied **real-time system design** using WebSocket and push notification services.

SELECTED SYSTEMS & PROJECTS

eKYC Platform (Digital Account Opening System)
.NET Core, React, Tailwind, Python, FastAPI

- Designed full-stack system for digital onboarding with multi-stage approval workflows.
- Integrated **AI-based verification pipeline**:
  - NID OCR and face matching (Polygon API)
  - Document OCR, address extraction, and profession prediction (Google APIs)
- Implemented AML checks, deduplication, and transaction behavior capture.
- Built real-time communication system using WebSocket and Firebase.

Bancassurance Platform
Node.js, Kafka, MongoDB, Netflix Conductor

- Designed distributed workflow system integrating multiple insurance providers.
- Implemented event-driven architecture for asynchronous processing.
- Developed fault-tolerant callback and retry mechanisms.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C/C++
ML/AI: PyTorch, TensorFlow, NLP, Computer Vision, OCR systems
Systems: Linux, Distributed Systems, Kafka, REST APIs
Backend: Node.js, FastAPI, ASP.NET Core
Databases: MongoDB, PostgreSQL
Tools: Docker, Kubernetes, Git

HONORS & AWARDS

- RUET Merit Scholarship (Top 30 students) – 2020, 2021, 2022
- Leadership Award: Class Representative – 2020–2023